ASAP CRN Cloud Release

Major Release DOI: 10.5281/zenodo.8384742

README - Release v3.0.0

Overview

The ASAP CRN Cloud is pleased to announce the 3.0.0 release of data which expands our datasets and harmonized collections of Human Postmortem-derived Brain Sequencing (PMDBS), and also introduces non-human (Mouse) datasets. In addition to an update to the **PMDBS Single Cell (sc) RNAseq Collection**, and the **PMDBS bulk RNAseq Collection**, we are introducing two spatial RNAseq Collections – a **PMDBS Spatial RNAseq Collection** and a **Mouse Spatial RNASeq Collection** – derived from brain slices. Our other datasets are organized as **Other PMDBS** and **Other Mouse** Datasets. In future releases we will also make additional Collections and Datasets from different modalities available.

Collections

- **PMDBS scRNAseq Collection, v3.0.0 - DOI: 10.5281/zenodo.16979638**
- **PMDBS bulk RNAseq Collection, v1.1.0 - DOI: 10.5281/zenodo.16975686**
- **PMDBS spatial RNAseq Collection, v1.0.0 - DOI: 10.5281/zenodo.16979475**
- **Mouse Spatial RNAseq Collection, v1.0.0 - DOI: 10.5281/zenodo.16979297**

ASAP Teams: Team Hafler, Team Lee, Team Jakobsson, Team Scherzer, Team Hardy, Team Wood, Team Biederer, Team Cragg, Team Edwards
See CRN Cloud Authorship List for the full list of contributing investigators/researchers by dataset.
ASAP CRN GitHub: github.com/ASAP-CRN
Data Release Date: 2025-09-30
ASAP CRN Cloud Release Number: v3.0.0
ASAP CRN Release DOI: 10.5281/zenodo.16975257

Curated Data File Manifest: ASAP CRN Cloud File Manifest - v3.1

Datasets

PMDBS scRNAseq

Datasets curated with the `pmdbs_sc_rnaseq` workflow:

- `team-hafler-pmdbs-sn-rnaseq-pfc`
- `team-lee-pmdbs-sn-rnaseq`
- `team-hardy-pmdbs-sn-rnaseq`
- `team-scherzer-pmdbs-sn-rnaseq-mtg`
- `team-jakobsson-pmdbs-sn-rnaseq`
- `cohort-pmdbs-sc-rnaseq`

PMDBS bulk RNAseq

Datasets curated with the `pmdbs_bulk_rnaseq` workflow:

- `team-hardy-pmdbs-bulk-rnaseq`
- `team-lee-pmdbs-bulk-rnaseq-mfg`
- `team-wood-pmdbs-bulk-rnaseq`
- `cohort-pmdbs-bulk-rnaseq`

PMDBS Spatial RNAseq

Dataset curated with the `spatial_geomx` workflow

- `edwards-pmdbs-spatial-geomx-th`

Mouse Spatial RNAseq

Dataset curated with the `spatial_visium` workflow

- `cragg-mouse-spatial-visium-striatum`

Other PMDBS Data

Raw data from two PMDBS datasets are made available. The `team-jakobsson-pmdbs-bulk-rnaseq` dataset will be integrated into the PMDBS bulk RNAseq Collection in a future release.

- `team-scherzer-pmdbs-sn-rnaseq-mtg-hybsel`
- `team-jakobsson-pmdbs-bulk-rnaseq`

Other Mouse Data

Raw data from two Mouse Single-Cell (Nucleus) RNAseq datasets are available, and will be curated with a CRN Cloud Mouse scRNAseq workflow introduced in a future release.

- `team-biederer-mouse-sc-rnaseq`
- `team-cragg-mouse-sn-rnaseq-striatum`

Collection Details

PMDBS scRNAseq Collection **10.5281/zenodo.16979638**

Workflow Release - v3.0.0

- Release date: 2025-09-30
- GitHub release link
- Pipeline version: v3.0.0

PMDBS bulk RNAseq Collection **10.5281/zenodo.16975686**

Workflow Release - v1.1.0

- Release date: 2025-09-30
- GitHub release link

- Pipeline version: v1.1.0

PMDBS Spatial RNAseq Collection **10.5281/zenodo.16979475**

Workflow Release - v1.0.0

- Release date: 2025-09-30
- GitHub release link
- Pipeline version: v1.0.0

Mouse Spatial RNAseq Collection **10.5281/zenodo.16979297**

Workflow Release - v1.0.0

- Release date: 2025-09-30
- GitHub release link
- Pipeline version: v1.0.0

Raw Data

The *raw* data refers to the fastq files transferred by each ASAP CRN Team. These data are available with the caveat that they are stored in "requester pays" buckets on the google cloud platform. A google cloud billing account will need to be registered to access these data. More information is available HERE. gs://`asap-raw-<team_name>-<source>-<dataset_name>`

Metadata & Data Dictionary

The ASAP CRN Common Data Elements (CDE - v3.2) outlines the variables which are harmonized across the ASAP CRN. These metadata enable the harmonization of the data submitted by multiple teams. The metadata consists of five tables, which are described below.

Metadata refers to descriptive information about the overall study, individual samples, all protocols, and references to processed and raw data file names. Metadata was aggregated using the table-based submission for each team's dataset. The tables - STUDY, PROTOCOL, SUBJECT, SAMPLE, DATA ,etc. - can be thought of as spreadsheets, and each table is prepared as simple .csv files which constitute the tables outlined in the ASAP CRN CDE (final

sheet). These have been aggregated across submissions and unique ASAP IDs generated for each dataset, team, subject and sample. I.e. ASAP_dataset_id, ASAP_team_id, ASAP_subject_id, and ASAP_sample_id.

The metadata can be browsed on ASAP CRN Cloud *Explorer* and exposed in Verily Workbench.

This document.